



Irene Rodríguez Sánchez

PHYSICS APPLIED TO SIMULATION AND MODELING

About me

I am driven by deep learning and challenges that require analytical rigor and teamwork. My goal is to contribute to projects with a direct impact on the scientific community, bringing a proactive attitude and the ability to transform abstract physical models into tangible solutions.

Curiosity has always driven me to question the underlying mechanisms of nature. This motivation inspires me to expand my knowledge in the field of **biophysics** and the **modeling of biological systems**, combining **computational** tools, hand in hand with **research**.

Relevant Experience

Design and Modeling of an auditory software - Internship (2025)

Institute of Neuroscience of Castilla y León (INCyL)

- Electroacoustic **calibration** and verification.
- **Editing of signals** and sounds.
- **Design and execution of experiments** in psychoacoustics and human auditory electrophysiology.
- Participation in experiments in psychoacoustics and human auditory electrophysiology.
- **Data analysis**.
- **Design and development of software** for human auditory evaluation.
- **Computer simulation** of hearing and auditory processes.

IV International Conference of Young Researchers INNOVA (2025)

International Centre for Spanish, Salamanca, Spain

- **Poster communication**: "Simulación espacio-temporal del potencial de acción en células cardíacas".

Conference of Female Researchers from Castile and León (2026)

University of biology, León, Spain

- **Poster communication**: "Simulación de señales electrocardiográficas mediante el modelo de FitzHugh-Nagumo en tejidos bidimensionales".

Contact



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Salamanca, Spain

Technical Skills

Computational

Python

Fortran

MATLAB

Mathematica

NumPy

SciPy

Matplotlib

Jupyter

Data & Tools

SQL NoSQL

LaTeX

Git

Data Analytics

Signal Processing

Languages

Spanish:
Native

English:
B2 Cambridge

Skills

Numerical Simulation
Data Analysis
Analytical Thinking
Scientific communication

Projects



github.com/Irenne04



[Vocal Biometrics Analysis and Signal Processing](#)



[Análisis de Datos de MREID](#)



[TFG: "Estudio de la dinámica de propagación del potencial de acción en el tejido cardíaco mediante el modelo de FitzHugh-Nagumo"](#)



[Tinnitus Project](#)

Education

Master's degree in physics of complex systems and biophysics (2026-2027)

Master's degree from the University of Barcelona (UB)

Degree in Physics (2021-2026)



Physics degree from the University of Salamanca (USAL)

- [TFG](#): "Estudio de la dinámica de propagación del potencial de acción en el tejido cardíaco mediante el modelo de FitzHugh-Nagumo".

Course on "Dynamical Modeling Methods for Systems Biology" (2024)

Online course by Icahn School of Medicine at Mount Sinai

- Stochastic models and models based on ordinary and partial differential equations.

Course on "Integrated Analysis in Systems Biology" (2024)

Online course by the Icahn School of Medicine at Mount Sinai

- Experimental design and computational analysis of ohmic data using dynamic models, graph theory and PCA.

Course on "Introducción al Deep Learning" (2025)

Face-to-face course issued by the LUMES Research Center

- Neural network modeling: from classical architectures to Deep Learning and biologically inspired Convolutional Networks.

Course on "Data Analytics" (2026)

Blended learning course by the School of Industrial Organization (EOI)

- SQL/NoSQL, Data Warehouse, advanced visualization and dashboard development.